



PowerUp 2026

Outage Identification from Electricity Market Data: Quickest Change Detection Approach

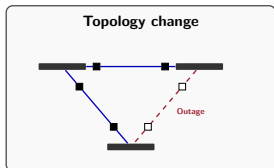
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Why topology information has value?

A market participant may update its **bidding decision** upon receiving new **outage information**.



Bidding decision

$$b_t^* = \arg \max_{b_t \in \mathcal{B}} \Pi_t(b_t; \mathcal{G}_t)$$

b_t : bidding decision at hour t

$\mathcal{G}_t$: assumed grid topology at hour t

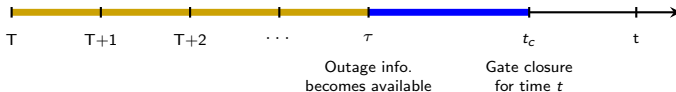
Π_t : profit function at hour t

Transmission
outage occurs

Detection window

Action window

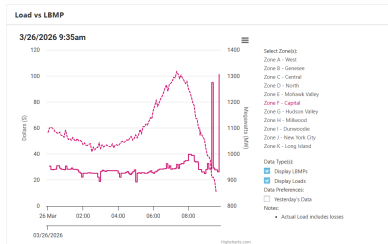
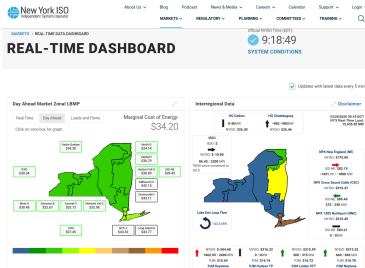
Operating hour



Takeaway: If the outage information becomes available before t_c , the market participant can update the next market decision.

How are unplanned outages communicated?

- ▶ Physical measurements (e.g., PMUs) enabling direct detection are **not accessible** to market participants.
- ▶ System operator reports may be **delayed, aggregated, or access-restricted**.
- ▶ **Opportunity:** **Public** market data enable **independent** outage detection.



NYISO Dashboard

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Research Question

How to design a **sequential detector** that identifies outages using publicly available **stream of market data**?

From Outage Signatures to Sequential Detection

Key Insight: From Physical Outages to Market Signals

A **topology change** induces a measurable and persistent **distributional shift** in LMPs, which can be exploited as a statistical signal for outage detection.

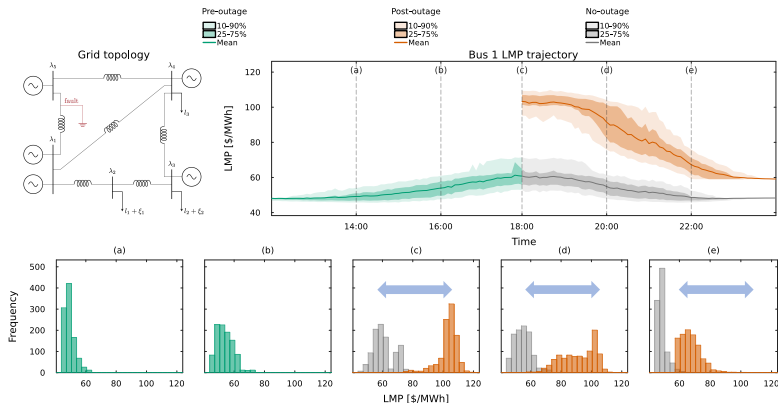


Figure: Topology-induced distributional shift in Bus 1 LMP under a line outage.

Problem Setup

- Market participants observe a **sequence** of LMP increments $\{\Delta\lambda_t\}$:

$$\Delta\lambda_t = \lambda_t - \lambda_{t-1}, \quad t = 1, 2, \dots$$

- We sequentially test the **nominal topology** against possible **outages**:

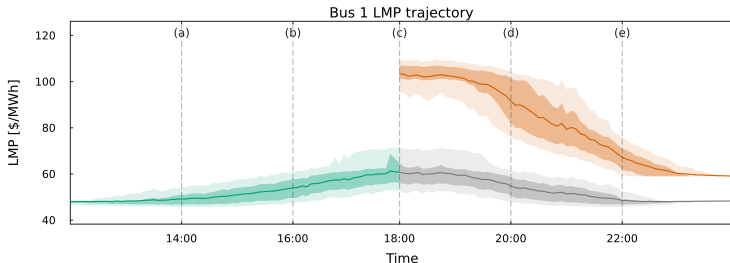
H_0 : Nominal operation

$H_{1,a}$: Outage “a”

$$\Delta\lambda_t \sim f_t^0, \quad \forall t \geq 1$$

$$\Delta\lambda_t \sim \begin{cases} f_t^0, & t < T, \\ f_t^a, & t \geq T, \end{cases} \quad a \in \mathcal{A}$$

- T is the **unknown change point**, and $a \in \mathcal{A}$ identifies the outage.



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Outputs

- (1) Detect the change as quickly as possible,
- (2) Identify which outage $a \in \mathcal{A}$ occurred.

Sequential Detection via Cumulative Sum (CuSum)

► Instantaneous evidence (log-likelihood ratio):

$$\ell_t = \log \frac{f_t^a(\Delta \lambda_t)}{f_t^0(\Delta \lambda_t)} \begin{cases} \rightarrow \mathbb{E}_a[\ell_t] = D_{\text{KL}}(f_t^a \| f_t^0) > 0 & \text{post-outage} \\ \rightarrow \mathbb{E}_0[\ell_t] = -D_{\text{KL}}(f_t^0 \| f_t^a) < 0 & \text{nominal} \end{cases}$$

► Accumulate evidence over time:

$$W_t = \max(0, W_{t-1} + \ell_t), \quad W_0 = 0$$

► Detection rule:

$$\tau = \inf\{t \geq 1 : W_t \geq \eta\}$$

Key challenge: How can we construct the distributions f_t^0 and f_t^a ?

Probabilistic Model of Market-Clearing Problem

Perturbed Market-Clearing Problem

Introduce demand uncertainty as a perturbation:

$$\mathbf{l} \rightarrow \mathbf{l} + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

$$\min_{\mathbf{p}, \boldsymbol{\ell}} \underbrace{\frac{1}{2} \mathbf{p}^\top \mathbf{C} \mathbf{p} + \mathbf{c}^\top \mathbf{p}}_{\text{generation cost}} + \underbrace{\frac{1}{2} \boldsymbol{\ell}^\top \mathbf{S} \boldsymbol{\ell} + \mathbf{s}^\top \boldsymbol{\ell}}_{\text{load-shedding cost}}$$

$$\text{s.t.} \quad \mathbf{1}^\top (\mathbf{M}_p \mathbf{p} - \mathbf{M}_l (\mathbf{l} + \boldsymbol{\xi} - \boldsymbol{\ell})) = 0 \quad \text{power balance}$$

$$|\mathbf{F}(\mathbf{M}_p \mathbf{p} - \mathbf{M}_l (\mathbf{l} + \boldsymbol{\xi} - \boldsymbol{\ell}))| \leq \bar{\mathbf{f}} \quad \text{transmission limits}$$

$$\underline{\mathbf{p}} \leq \mathbf{p} \leq \bar{\mathbf{p}} \quad \text{generation limits}$$

$$\mathbf{0} \leq \boldsymbol{\ell} \leq \mathbf{l} + \boldsymbol{\xi} \quad \text{load shedding}$$

Variables

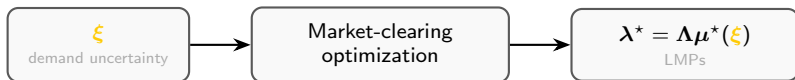
$\mathbf{p}(\boldsymbol{\xi})$	generation
$\boldsymbol{\ell}(\boldsymbol{\xi})$	load shedding
$\boldsymbol{\mu}(\boldsymbol{\xi})$	dual variable
$\boldsymbol{\lambda}(\boldsymbol{\xi}) = \boldsymbol{\Lambda} \boldsymbol{\mu}(\boldsymbol{\xi})$	LMPs

Network model

$\mathbf{F}$	PTDF matrix
$\bar{\mathbf{f}}$	line limits
$\mathbf{M}_p$	generator mapping
$\mathbf{M}_l$	load mapping

The auxiliary matrix to compute LMPs from the dual variables: $\boldsymbol{\Lambda} = \begin{bmatrix} \mathbf{1} & -\mathbf{F}^\top & \mathbf{F}^\top & \mathbf{0} \end{bmatrix}$

Mapping from Demand Uncertainty to Market Outcomes



For each ξ , the market-clearing problem induces the **optimization mapping**

$$\xi \mapsto \lambda^* = \mathcal{M}(\xi).$$

Goal: Given f_ξ , construct the induced density f_λ .

- ▶ $\mathcal{M}(\cdot)$ is defined **implicitly** through an optimization problem.
- ▶ Generic density transformation may require **inverse** and the **Jacobian** of $\mathcal{M}$.

Challenge: How can we make this optimization-induced mapping **explicit** and **tractable**?

Multi-Parametric Programming: Making the Mapping Explicit

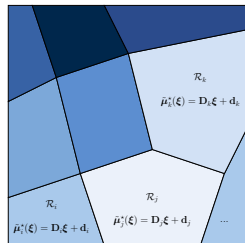
Idea: Globally implicit optimization mapping becomes **piecewise affine and explicit**.

Parametric QP Form

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{q}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} \leq \mathbf{B} \boldsymbol{\xi} + \mathbf{b} : \boldsymbol{\mu}. \end{aligned}$$

$$\mathbf{x} = [\mathbf{p}^\top, \boldsymbol{\ell}^\top]^\top$$

MPP $\rightarrow$



Each region $\mathcal{R}_r$ corresponds to a fixed active set.

— Ji, Yuting and Thomas, Robert J. and Tong, Lang, "Probabilistic Forecasting of Real-Time LMP and Network Congestion," IEEE Trans. Power Systems, 2017.

— $\bar{\mathbf{A}}, \bar{\mathbf{B}}, \bar{\mathbf{b}}$ contain those entries of $\mathbf{A}, \mathbf{B}, \mathbf{b}$ that correspond to the active constraints.

Multi-Parametric Programming: Making the Mapping Explicit

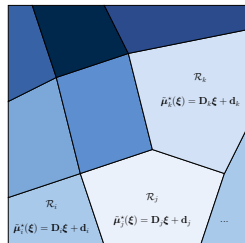
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MPP $\rightarrow$



Each region $\mathcal{R}_r$ corresponds to a fixed active set.

For a fixed set of **active constraints**, the KKT conditions are **linear** in $\hat{\boldsymbol{\xi}}$:

$$\tilde{\boldsymbol{\mu}}^*(\hat{\boldsymbol{\xi}}) = \mathbf{D} \hat{\boldsymbol{\xi}} + \mathbf{d}.$$

$$\mathbf{D} = -(\tilde{\mathbf{A}} \mathbf{Q}^{-1} \tilde{\mathbf{A}}^\top)^{-1} \tilde{\mathbf{B}},$$

$$\mathbf{d} = -(\tilde{\mathbf{A}} \mathbf{Q}^{-1} \tilde{\mathbf{A}}^\top)^{-1} (\tilde{\mathbf{b}} + \tilde{\mathbf{A}} \mathbf{Q}^{-1} \mathbf{q}).$$

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Construction of Critical Regions

- ▶ This solution is valid only when the same constraints remain **active**.
- ▶ This defines a **critical region** $\mathcal{R}_i$:
 - ▶ inactive constraints remain **strictly feasible**
 - ▶ active constraints retain **nonnegative dual variables**

$$\mathcal{R}_i = \left\{ \xi \mid (\bar{\mathbf{A}}\mathbf{P}_i - \bar{\mathbf{B}})\xi < \bar{\mathbf{b}} - \bar{\mathbf{A}}\mathbf{p}_i, \mathbf{D}_i\xi + \mathbf{d}_i \geqslant \mathbf{0} \right\}$$

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Algorithm: Constructing the Critical Regions

Input: Parametric QP and uncertainty set Ξ

Output: Critical regions $\mathcal{CR}$

- 1: Initialize $\mathcal{CR} \leftarrow \emptyset$
- 2: **while** there are uncovered points in Ξ **do**
- 3: Select an uncovered point $\hat{\xi} \in \Xi$
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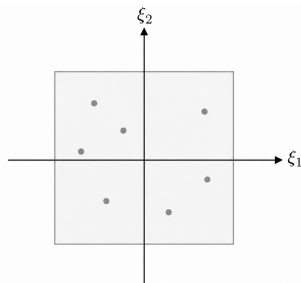
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Start with the 2D uncertainty space

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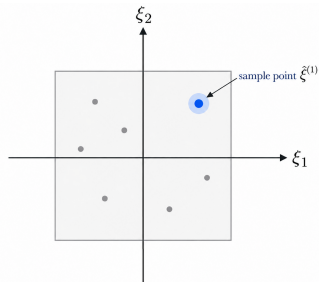
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Start with the 2D uncertainty space

Solve the QP at the sampled point

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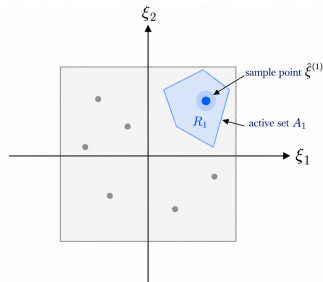
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Construct the first critical region
The active set stays unchanged inside R_1

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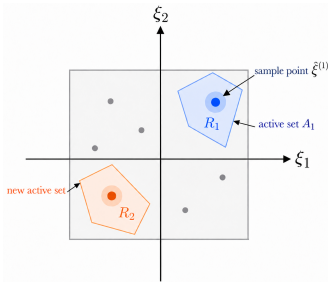
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Find an uncovered sample point
A different active set defines a new region

From Critical Regions to Probability Distributions

- ▶ LMPs follow as an affine mapping:

$$\lambda_i(\xi) = \tilde{\mathbf{A}}_i \mathbf{D}_i \xi + \tilde{\mathbf{A}}_i \mathbf{d}_i$$

- ▶ Consider LMP increments:

$$\Delta \lambda_t = \lambda_i(\xi_t) - \lambda_i(\xi_{t-1}) = \tilde{\mathbf{A}}_i \mathbf{D}_i (\xi_t - \xi_{t-1})$$

- ▶ Since $\xi_t - \xi_{t-1} \sim \mathcal{N}(0, \Sigma)$:

$$\mathbb{E}[\Delta \lambda_t] = \mathbf{0}, \quad \text{Cov}[\Delta \lambda_t] = \tilde{\mathbf{A}}_i \mathbf{D}_i \Sigma (\tilde{\mathbf{A}}_i \mathbf{D}_i)^\top$$

- ▶ Within each region $\mathcal{R}_i$, the **likelihood** for $f_i(\Delta \lambda_t)$ for **CuSum** detection is

$$\Delta \lambda_t \sim \mathcal{N}(\mathbf{0}, \Sigma_i), \quad \Sigma_i = \tilde{\mathbf{A}}_i \mathbf{D}_i \Sigma (\tilde{\mathbf{A}}_i \mathbf{D}_i)^\top$$
$$f_i(\Delta \lambda_t) = \frac{1}{(2\pi)^{n/2}} \frac{1}{|\Sigma_i|^{1/2}} \exp\left(-\frac{1}{2} \Delta \lambda_t^\top \Sigma_i^{-1} \Delta \lambda_t\right).$$

Constructing the CuSum Statistic

- Compare nominal operation vs outage scenario $a \in \mathcal{A}$:

$$f_{i(t)}^0 : \mathcal{N}(\mathbf{0}, \Sigma_i^0), \quad f_{j(t)}^a : \mathcal{N}(\mathbf{0}, \Sigma_j^a)$$

- Log-likelihood ratio (instantaneous evidence):

$$\ell_t^{(a)} = \log \left(\frac{f_j^a(\Delta \lambda_t)}{f_i^0(\Delta \lambda_t)} \right) = \frac{1}{2} \log \left(\frac{|\Sigma_i^0|}{|\Sigma_j^a|} \right) + \frac{1}{2} \Delta \lambda_t^\top \left[(\Sigma_i^0)^{-1} - (\Sigma_j^a)^{-1} \right] \Delta \lambda_t.$$

- Accumulate evidence via CuSum:

$$W_t^{(a)} = \max(0, W_{t-1}^{(a)} + \ell_t^{(a)}), \quad \forall a \in \mathcal{A}.$$

Threshold Selection

- ▶ Run CuSum statistics in **parallel** for all outage candidates $a \in \mathcal{A}$.
- ▶ Detection and identification rules:

$$\underbrace{\tau}_{\text{alarm time}} = \inf \left\{ t \geq 1 : \max_{a \in \mathcal{A}} \overbrace{W_t^{(a)}}^{\text{CuSum}} \geq \underbrace{\eta}_{\text{threshold}} \right\}$$
$$\hat{a} = \arg \max_{a \in \mathcal{A}} W_{\tau}^{(a)}$$

- ▶ Choose η such that **Average Run Length** (ARL) meets a desired target:

$$\text{ARL}_0 = \mathbb{E}_{\mathbb{H}_0}[\tau]$$

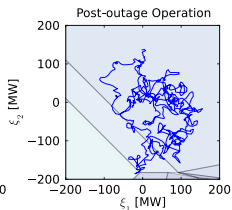
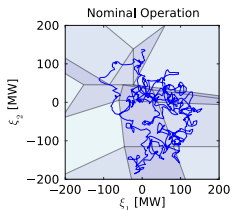
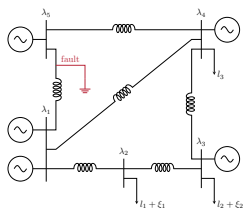
- ▶ Expected detection delay in post-outage operation (performance metric):

$$\mathbb{E}_{\mathbb{H}_1}[\tau - T \mid \tau \geq T]$$

Results

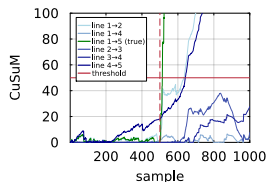
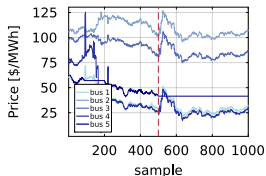
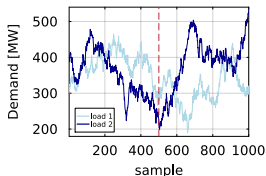
Experiment Setup

- Fault on the line connecting buses 1 and 5 at $T = 500$.
- The outage alters the **multi-parametric geometry** of the market-clearing problem for the **nominal** and the **post-outage** market operation.
- PJM system and the corresponding critical regions:



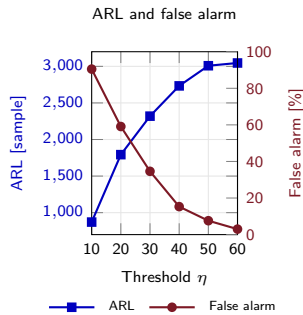
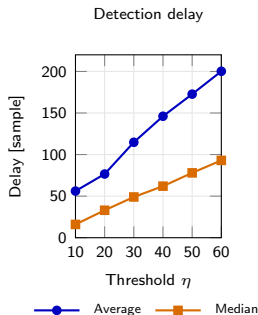
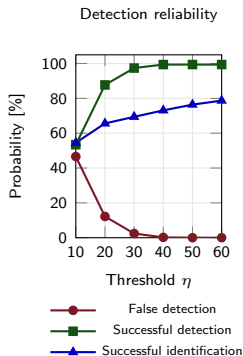
Distinguishing Line Outages from Demand Variation

- ▶ The outage at $T = 500$ coincides with a sudden demand growth.
- ▶ Demand and price trajectories might mistakenly attribute the price spike to demand growth rather than the outage.
- ▶ Demand (left) and LMP (middle) data streams and the dynamics of the six CuSum statistics (right). The outage occurs at $T = 500$ (red dashed line):



Performance Analysis of the CuSum Statistic

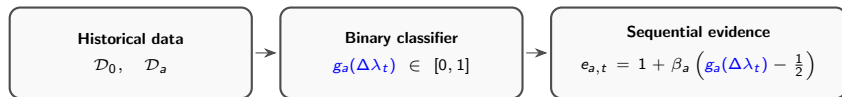
- ▶ Increasing the threshold η improves **detection reliability**, but at the cost of a longer **detection delay**.
- ▶ Higher thresholds increase **ARL** and reduce the **false-alarm probability**.



- ▶ Developed a **sequential detector** for power systems outages using **stream** of real-time LMP data
- ▶ Leveraged **multi-parametric programming** to obtain tractable **probabilistic models** of LMPs
- ▶ Controlled **false alarms** through ARL-based threshold calibration
- ▶ Future work: Develop classifier-based **non-parametric** detector under model **misspecification**.

Future work: Classifier-based non-parametric detection

- ▶ MPP-based CuSum enables detection when the **system parameters** and induced observation densities are accurately specified.
- ▶ We develop a **non-parametric** detector that learns outage-discriminating information directly from historical observations.



- ▶ Accumulate evidence through the **e-process**

$$W_{a,t} = \prod_{s=1}^t e_{a,s}.$$

- ▶ Given a threshold $\eta > 0$, stopping time is

$$\tau = \inf \left\{ t \geq 1 : \max_{a \in \mathcal{A}} W_{a,t} \geq \eta \right\}.$$

Thank You!



Google Scholar Profile